



A study of the quasi covering dimension of finite lattices

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Abstract

The notion of dimension for posets has been studied extensively (see, for example, Bae in J Korean Math Soc 36(1):15–36, 1999, Dube et al. in Discr Math 338:1096–1110, 2015, Gierz et al. in A compendium of continuous lattices. Springer, Berlin, 1980, Trotter in Combinatorics and partially ordered sets: dimension theory. Johns Hopkins University Press, Baltimore, 1992, Vinokurov in Soviet Math Dokl 168(3):663–666, 1966, Zhang et al. in Discr Math 340(5):1086–1091, 2017). Moreover, in the view of matrix theory, some of these dimensions, such as the order dimension, the Krull dimension and the covering dimension, have been studied using the so-called order and incidence matrices (see Boyadzhiev et al. in Appl Math Comput 333:276–285, 2018, Dube et al. in Filomat 31(10):2901–2915, 2017, Georgiou et al. in Quaest Math 39(6):797–814, 2016). In this paper, we define a new dimension for finite lattices, called the quasi covering dimension, and we study many of its properties. We characterize it using matrices and we present an algorithm for computing this dimension.

Keywords Quasi covering dimension · Quasi cover · Finite lattice · Dense element · Matrix theory

Mathematics Subject Classification 06A07

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1 Introduction

The dimension theory of frames (especially, their covering dimension, their small inductive dimension and their large inductive dimension) attracted the interest of many researchers such as Charalambous, Gilmour, Banaschewski, Brijlall, Baboolal (see Banaschewski and Gilmour 1989; Brijlall and Baboolal 2008, 2010; Charalambous 1974).

Recently, this theory was enriched by the study in Georgiou et al. (2019), where a new dimension for frames, which is called the quasi covering dimension, was defined and many of its properties were investigated.

In this paper, we develop a theory for the quasi covering dimension of finite lattices, proving many results.

Moreover, it is known that finite posets and studies of their dimensions have important roles in discrete mathematics. This role is based on matrix algebra and algorithms (see Yáñez and Montero 1999). For example, the order dimension, the Krull dimension and the covering dimension have been studied using matrices (see Boyadzhiev et al. 2018; Dube et al. 2017; Georgiou et al. 2016).

Therefore, we complete our paper by studying the quasi covering dimension of finite lattices in the view of matrix algebra. Both order matrices and incidence matrices play important roles in this study.

Especially, in Sect. 2, we remind basic definitions and notations from dimension theory of finite lattices and matrix theory. In Sect. 3, we define the quasi covering dimension for finite lattices and in Sect. 4, we compare several of its properties. In Sect. 5, since the meaning of the dense element of an arbitrary finite lattice will play an essential role in this study, we characterize it in terms of incidence matrices. Finally, in Sect. 6, we present an algorithm for computing the quasi covering dimension of an arbitrary finite lattice.

2 Preliminaries

In this section we recall some definitions and notations that are needed in the rest of our study. We mention that in what follows, by $X = \{x_1, \dots, x_n\}$ we mean a finite lattice $(X, \leq)$ for which $x_1 = 0_X$ and $x_n = 1_X$ are the least and the greatest element of X , respectively. Additionally, for an element x of X , we use the notation of $\downarrow x = \{y \in X : y \leq x\}$.

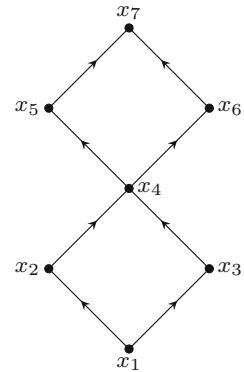
Definition 1 Let X be a finite lattice. A subset C of X is called a *cover* of X if $\bigvee C = x_n$ and $x_1 \notin C$. A cover D of X is called a *refinement* of a cover C of X if for each $d \in D$ there exists $c \in C$ such that $d \leq c$.

Definition 2 Let X be a finite lattice. The *order* of a subset C of X , $\text{ord}(C)$, is defined to be k , where $k \in \{0, 1, \dots\}$, if and only if the infimum of any $k + 2$ distinct elements of C is x_1 and there exist $k + 1$ distinct elements of C whose infimum is not x_1 .

Definition 3 The function $\dim$, called *covering dimension*, with domain the class of all finite lattices and range the set $\{0, 1, \dots\}$, is defined as follows:

- (1) $\dim(X) \leq k$, where $k \in \{0, 1, \dots\}$, if and only if for every cover C of X , there exists a cover R of X such that R is a refinement of C and $\text{ord}(R) \leq k$.
- (2) $\dim(X) = k$, where $k \in \{1, 2, \dots\}$, if $\dim(X) \leq k$ and $\dim(X) \not\leq k - 1$.

Fig. 1 The finite lattice $(X, \leq)$



Definition 4 Let $(X, \leq)$ be a finite lattice. The $n \times n$ matrix $T_X^{\leq} = (t_{ij})$, where

$$t_{ij} = \begin{cases} 1, & \text{if } x_i \leq x_j \\ 0, & \text{otherwise} \end{cases}$$

is called the *incidence matrix* of X . The n columns of the incidence matrix $T_X^{\leq}$ are denoted by $c_1(T_X^{\leq}), \dots, c_n(T_X^{\leq})$. Furthermore, in what follows, by the notion of $\max(c_i(T_X^{\leq}) + c_j(T_X^{\leq}))$, where $i, j \in \{1, \dots, n\}$, we shall mean the maximum element of the matrix $c_i(T_X^{\leq}) + c_j(T_X^{\leq})$.

Definition 5 Let $(X, \leq)$ be a finite lattice. The $n \times n$ matrix $A_X^{\leq} = (a_{ij})$, where

$$a_{ij} = \begin{cases} 1, & \text{if } i = j \\ 2, & \text{if } x_i < x_j \\ -2, & \text{if } x_j < x_i \\ 0, & \text{if } x_i \parallel x_j \end{cases}$$

is called the *order matrix* of X . The n rows of the order matrix $A_X^{\leq}$ are denoted by $r_1(A_X^{\leq}), \dots, r_n(A_X^{\leq})$. The set of all elements of the order matrix $A_X^{\leq}$ is denoted by $\mathcal{E}(A_X^{\leq})$. We state that for the simplicity of the writing if α is an element of a matrix A (respectively, α is not an element of a matrix A) we shall write $\alpha \in A$ (respectively, $\alpha \notin A$).

Example 1 Let $(X, \leq)$ be the finite lattice represented in Fig. 1.

Then the incidence matrix $T_X^{\leq}$ and the order matrix $A_X^{\leq}$ of X are given, as follows, respectively:

$$T_X^{\leq} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad A_X^{\leq} = \begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 2 & 2 \\ -2 & 1 & 0 & 2 & 2 & 2 & 2 \\ -2 & 0 & 1 & 2 & 2 & 2 & 2 \\ -2 & -2 & -2 & 1 & 2 & 2 & 2 \\ -2 & -2 & -2 & -2 & 1 & 0 & 2 \\ -2 & -2 & -2 & -2 & 0 & 1 & 2 \\ -2 & -2 & -2 & -2 & -2 & -2 & 1 \end{pmatrix}.$$

3 The quasi covering dimension of finite lattices

In this section, we define the quasi covering dimension of an arbitrary finite lattice and we present basic of its properties.

Definition 6 (Gierz et al. 1980) Let X be a finite lattice. A subset C of X is called a *quasi cover* of X if $\bigvee C = x$, where x is a dense element of X (that is, $x \wedge y \neq 0_X$, for every $y \in X \setminus \{0_X\}$). Two quasi covers C_1 and C_2 of X are called *similar* (in short, we write $C_1 \sim C_2$) if $\bigvee C_1 = \bigvee C_2$. A quasi cover D of X is called a *refinement* of a quasi cover C of X if for each $d \in D$ there exists $c \in C$ such that $d \leq c$.

Definition 7 The function $\dim_q$, called *quasi covering dimension*, with domain the class of all finite lattices and range the set $\{0, 1, \dots\}$, is defined as follows:

- (1) $\dim_q(X) \leq k$, where $k \in \{0, 1, \dots\}$, if and only if for every quasi cover C of X , there exists a quasi cover R of X such that $C \sim R$, R is a refinement of C and $\text{ord}(R) \leq k$.
- (2) $\dim_q(X) = k$, where $k \in \{1, 2, \dots\}$, if $\dim_q(X) \leq k$ and $\dim_q(X) \not\leq k - 1$.

Proposition 1 For every finite lattice X we have

$$\dim_q(X) = \max\{\dim(\downarrow x) : x \text{ is a dense element of } X\}.$$

Proof We shall prove the inequality

$$\dim_q(X) \leq \max\{\dim(\downarrow x) : x \text{ is a dense element of } X\}.$$

Let $\max\{\dim(\downarrow x) : x \text{ is a dense element of } X\} = k$, where $k \in \{0, 1, \dots\}$. We prove that $\dim_q(X) \leq k$. Let C be a quasi cover of X , that is $\bigvee C = x$, where x is a dense element of X . Since $1_{\downarrow x} = x$, the set C is a cover of the lattice $\downarrow x$. By assumption we have $\dim(\downarrow x) \leq k$. Thus, there exists a cover R of $\downarrow x$, which refines the cover C and satisfies the inequality $\text{ord}(R) \leq k$. Since $\bigvee R = x$, R is a quasi cover of X , which is similar to C . Therefore, $\dim_q(X) \leq k$.

Conversely, we shall prove the inequality

$$\max\{\dim(\downarrow x) : x \text{ is a dense element of } X\} \leq \dim_q(X).$$

We suppose that $\dim_q(X) = k$, where $k \in \{0, 1, \dots\}$, and we prove that the maximum value of $\{\dim(\downarrow x) : x \text{ is a dense element of } X\}$ is less than or equal to k . Let x_0 be a dense element of X . It suffices to prove that $\dim(\downarrow x_0) \leq k$. Let C be a cover of $\downarrow x_0$, that is $\bigvee C = x_0$. Since x_0 is a dense element of X , the set C is a quasi cover of X . Additionally, since $\dim_q(X) = k$, there exists a quasi cover R of X , which refines C , is similar to C and satisfies the relation $\text{ord}(R) \leq k$. Since, $R \sim C$, the set R is a cover of the lattice $\downarrow x_0$. Therefore, $\dim(\downarrow x_0) \leq k$. $\square$

Remark 1 (1) For every finite lattice X we have $\dim(X) \leq \dim_q(X)$.

(2) Isomorphic finite lattices have the same quasi covering dimension.

(3) If 1_X is the only dense element of X , then $\dim(X) = \dim_q(X)$.

Example 2 (1) For the finite lattices which are represented in Fig. 2 we have $\dim(X) = \dim_q(X) = 0$ and $\dim(Y) = 0$, $\dim_q(Y) = 1$.

(2) For the finite lattice which is represented in Fig. 3 we have $\dim(X) = 1$ and $\dim_q(X) = 2$.

Fig. 2 The finite lattices $(X, \leq_1)$ and $(Y, \leq_2)$

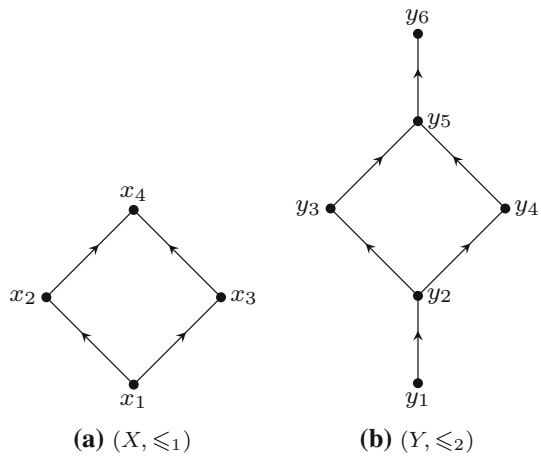
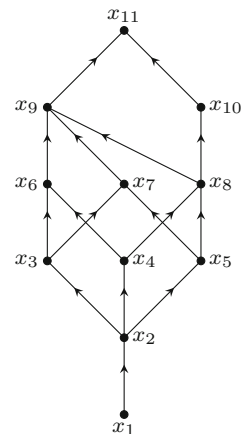


Fig. 3 The finite lattice $(X, \leq)$



Proposition 2 For any $k \in \{1, 2, \dots\}$, there exists a finite lattice $X(k)$ with $\dim_q(X(k)) = k$ and $\dim(X(k)) = 1$.

Proof Let $A_k = \{a_0, a_1, a_2, \dots, a_k\}$ and $\mathcal{P}(A_k)$ be the power set of A_k . We now consider the finite lattice $X(k) = \mathcal{P}(A_k) \cup \{b, 0_{X(k)}, 1_{X(k)}\}$ (see Fig. 4) which is endowed by the partial order $\leq$ for which:

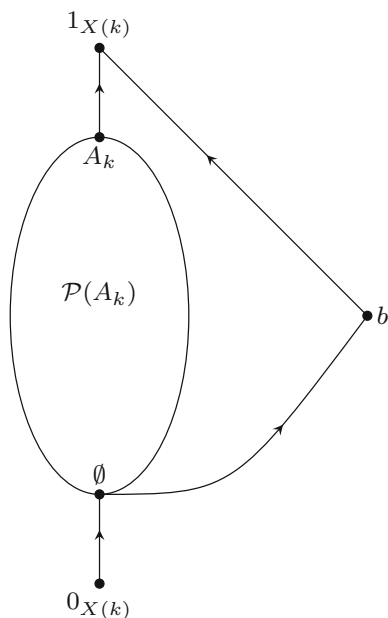
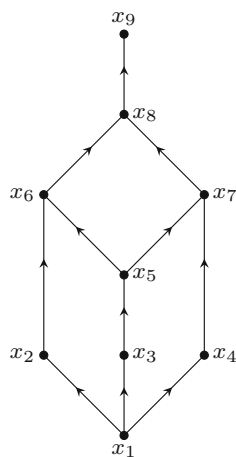
- (i) $0_{X(k)} \leq x \leq 1_{X(k)}$, for every $x \in X(k)$,
- (ii) $x \leq y \Leftrightarrow x \subseteq y$, for every $x, y \in \mathcal{P}(A_k)$,
- (iii) $x \parallel b$, for every $x \in X(k) \setminus \{\emptyset, b, 0_{X(k)}, 1_{X(k)}\}$.

Suppose that $k \in \{1, 2, \dots\}$. Since every cover of $X(k)$ contains b , $\dim(X(k)) = 1$. By Proposition 1 we have

$$\dim_q(X(k)) = \max\{\dim(\downarrow x) : x \text{ is dense element of } X(k)\} = \dim(\downarrow A_k) = k.$$

□

Remark 2 We observe that there exists a sublattice Y of a finite lattice X such that $\dim_q(Y) > \dim_q(X)$ as shown in Fig. 5.

Fig. 4 The lattice $(X(k), \leq)$ **Fig. 5** The finite lattice $(X, \leq)$ 

Indeed, we have $\dim_q(X) = 0$. But, for the sublattice $Y = \{x_3, x_5, x_6, x_7, x_8, x_9\}$ we have $\dim_q(Y) = 1$. Thus, we have the following proposition.

Proposition 3 *Let X be a finite lattice and Y be a sublattice of X . If every dense element of Y is also a dense element of X , then $\dim_q(Y) \leq \dim_q(X)$. Moreover, if Y and X have exactly the same dense elements, then $\dim_q(Y) = \dim_q(X)$.*

Proof Suppose that every dense element of Y is also a dense element of X . By Proposition 1 there exists a dense element y_0 of Y such that $\dim_q(Y) = \dim(\downarrow y_0)$. By assumption y_0 is a dense element of X . Therefore,

$$\dim_q(Y) = \dim(\downarrow y_0) \leq \max\{\dim(\downarrow x) : x \text{ is a dense element of } X\} = \dim_q(X).$$

Now, suppose that Y and X have the same dense elements. By Proposition 1 there exists a dense element x_0 of X such that $\dim_q(X) = \dim(\downarrow x_0)$. By assumption x_0 is a dense element of Y . Therefore,

$$\dim_q(X) = \dim(\downarrow x_0) \leq \max\{\dim(\downarrow y) : y \text{ is a dense element of } Y\} = \dim_q(Y).$$

□

We complete this section introducing a different characterization of the quasi covering dimension of an arbitrary finite lattice X .

Remark 3 The relation $\sim$ presenting the similarity of two quasi coverings is an equivalence relation on the finite lattice X . We denote by $qc(X, \sim)$ the collection of all equivalence classes under $\sim$. Every equivalence class is completely determined by a dense element x and is denoted by c_x , that is

$$c_x = \{\{x_{i_1}, \dots, x_{i_m}\} \subseteq X : x_{i_1} \vee \dots \vee x_{i_m} = x\}.$$

Definition 8 Let $c_x \in qc(X, \sim)$. A quasi cover $C \in c_x$ of X is called c_x -minimal cover if $C \subseteq R$ for every refinement R of C with $R \in c_x$.

Lemma 1 If X is a finite lattice and C is a quasi cover of X with $C \in c_x$ and $x \notin C$, then there exists a c_x -minimal cover R of X which refines C and $x \notin R$.

Proof We consider the set

$$C' = \{y \in C : \text{there does not exist } z \in C \text{ such that } y < z\}.$$

Then, $C' \in c_x$, C' is an antichain and $C' \subseteq C$. If C' is a c_x -minimal cover of X , then we are done. If C' is not a c_x -minimal cover of X , then there exists a refinement R_1 of C' such that $R_1 \in c_x$ and $C' \not\subseteq R_1$ (clearly, $x \notin R_1$). Without loss of generality, we can assume that R_1 is an antichain. Let $x_{i_1} \in C' \setminus R_1$. If R_1 is a c_x -minimal cover of X , then we are done. If R_1 is not a c_x -minimal cover of X , then there exists a refinement R_2 of R_1 such that $R_2 \in c_x$ and $R_1 \not\subseteq R_2$ (clearly, $x \notin R_2$). Let $x_{i_2} \in R_1 \setminus R_2$. Then, $x_{i_1} \neq x_{i_2}$. If R_2 is a c_x -minimal cover of X , then we are done. If R_2 is not a c_x -minimal cover of X , then there exists a refinement R_3 of R_2 such that $R_3 \in c_x$ and $R_2 \not\subseteq R_3$ (clearly, $x \notin R_3$). Let $x_{i_3} \in R_2 \setminus R_3$. Then, $x_{i_2} \neq x_{i_3}$. We shall prove that $x_{i_1} \neq x_{i_3}$. Let suppose that $x_{i_1} = x_{i_3}$. Since $x_{i_3} \in R_2$, $x_{i_1} \notin R_1$ and R_2 refines R_1 , there exists $r_1 \in R_1$ such that $x_{i_1} < r_1$. Moreover, since R_1 refines C' and $r_1 \in R_1$, there exists $c' \in C'$ such that $r_1 \leq c'$. Thus, $x_{i_1} < c'$, which is a contradiction since C' is an antichain. Therefore, $x_{i_1} \neq x_{i_3}$. Following this way, we produce an infinite sequence $x_{i_1}, x_{i_2}, x_{i_3}, \dots$ of distinct elements of X , which is a contradiction since X is a finite set. Thus, there exists a c_x -minimal cover R of X which refines C and $x \notin R$. □

Proposition 4 Let X be a finite lattice and $k \in \{0, 1, \dots\}$. Then, $\dim_q(X) \leq k$ if and only if for every $c_x \in qc(X, \sim)$, each c_x -minimal cover C of X satisfies the relation $\text{ord}(C) \leq k$.

Proof Let $\dim_q(X) \leq k$, $c_x \in qc(X, \sim)$ and C be a c_x -minimal cover of X . We prove that $\text{ord}(C) \leq k$. Since $\dim_q(X) \leq k$, there exists a refinement R of C which is similar to C and satisfies the relation $\text{ord}(R) \leq k$. If $R = C$, then we are done. If $R \neq C$, since C is a c_x -minimal cover of X and $R \in c_x$ which refines C , by Definition 8, we have $C \subseteq R$. Therefore, $\text{ord}(C) \leq k$.

Conversely, we prove that $\dim_q(X) \leq k$. Let C be a quasi cover of X . That is, $C \in c_x$, for some $c_x \in qc(X, \sim)$. Clearly, if $x \in C$, then $\{x\}$ is a quasi cover of X , which is a refinement

of C , is similar to C and $\text{ord}(\{x\}) = 0 \leq k$. If $x \notin C$, then, by Lemma 1, there exists a c_x -minimal cover R of X , which refines C and $x \notin R$. Therefore, by assumption $\text{ord}(R) \leq k$ and thus, $\dim_q(X) \leq k$. $\square$

Corollary 1 For every finite lattice X we have

$$\dim_q(X) = \max\{\text{ord}(C) : C \in c_x \in qc(X, \sim)\}.$$

Proof It follows directly by Proposition 4. $\square$

Proposition 5 Let $c_x \in qc(X, \sim)$. Then, every c_x -minimal cover of X is an antichain.

Proof Let C be a c_x -minimal cover of X . We prove that C is an antichain. In contrast, if there exist two distinct elements x_i and x_j of C such that $x_i \leq x_j$, then the set $R = C \setminus \{x_i\}$ is a quasi cover of X , which refines C , is similar to C and $R \in c_x$. Since C is a c_x -minimal cover of X , we have $C \subseteq R$, which is a contradiction. Therefore, C is an antichain. $\square$

4 Properties of the quasi covering dimension of finite lattices

In this section, we study certain properties of the quasi covering dimension of different finite lattices. Especially, having two finite lattices, we study the quasi covering dimension of their linear sum, Cartesian product, lexicographic product and finally, of their rectangular product.

Definition 9 (Davey and Priestley 2002) The *linear sum* $X \oplus Y$ of two partially ordered sets $(X, \leq_1)$ and $(Y, \leq_2)$, where $X \cap Y = \emptyset$, is the partially ordered set $(X \cup Y, \leq)$, where the relation $\leq$ is defined as follows:

$$x \leq y \Leftrightarrow \begin{cases} x, y \in X & \text{and } x \leq_1 y \\ x, y \in Y & \text{and } x \leq_2 y \\ x \in X, y \in Y. \end{cases}$$

Clearly, if $(X, \leq_1)$ and $(Y, \leq_2)$ are finite lattices, then $(X \oplus Y, \leq)$ is also a finite lattice.

Remark 4 (1) The example in Fig. 6 verifies the claim

$$\dim_q(X \oplus Y) \neq \dim_q(Y \oplus X).$$

Indeed, we have $\dim_q(X) = \dim_q(Y) = 0$. Furthermore, $\dim_q(Y \oplus X) = 0$ and $\dim_q(X \oplus Y) = 1$. Thus, $\dim_q(X \oplus Y) \neq \dim_q(Y \oplus X)$.

(2) The example in Fig. 7 verifies that the assertion

$$\dim_q(X \oplus Y) \leq \dim_q(X) + \dim_q(Y) + 1$$

does not hold for all the finite lattices X and Y .

Indeed, we have $\dim_q(X) = \dim_q(Y) = 0$ and $\dim_q(X \oplus Y) = 2$. Thus,

$$\dim_q(X \oplus Y) \not\leq \dim_q(X) + \dim_q(Y) + 1.$$

Definition 10 (Davey and Priestley 2002) The *Cartesian product* of two partially ordered sets $(X, \leq_1)$ and $(Y, \leq_2)$ is the partially ordered set $(X \times Y, \leq)$, where the relation $\leq$ is defined as follows:

$$(x_1, y_1) \leq (x_2, y_2) \Leftrightarrow x_1 \leq_1 x_2 \text{ and } y_1 \leq_2 y_2.$$

We state that if $(X, \leq_1)$ and $(Y, \leq_2)$ are finite lattices, then $(X \times Y, \leq)$ is also a finite lattice.

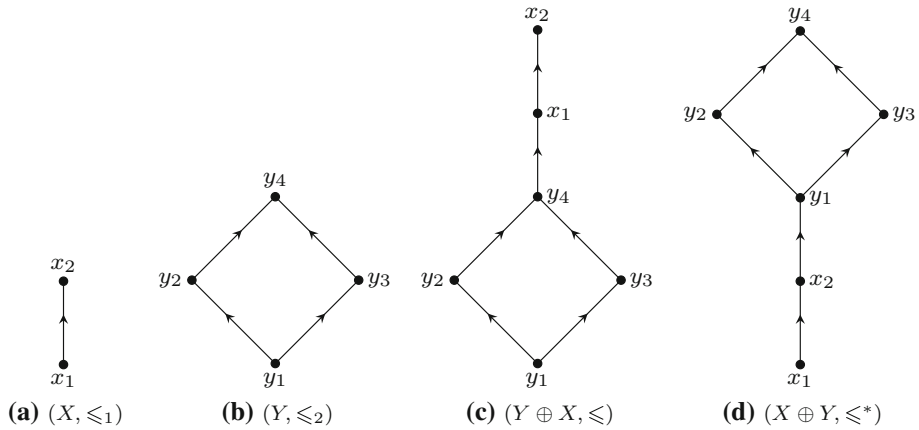


Fig. 6 The finite lattices $(X, \leq_1)$, $(Y, \leq_2)$, $(Y \oplus X, \leq)$ and $(X \oplus Y, \leq^*)$

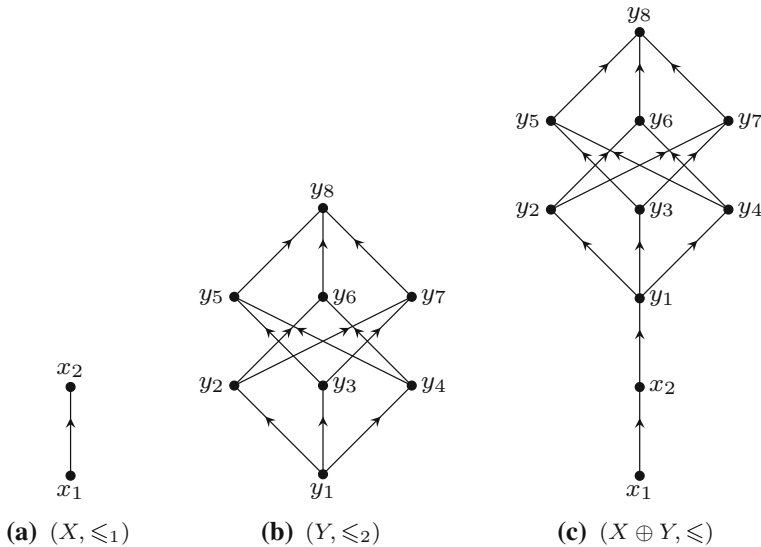


Fig. 7 The finite lattices $(X, \leq_1)$, $(Y, \leq_2)$ and $(X \oplus Y, \leq)$

To compute the quasi covering dimension of the Cartesian product $X \times Y$ of two finite lattices X and Y , we compare the following two results which can, easily, be proved.

Lemma 2 *Let $(x, y) \in X \times Y$. Then, $\downarrow(x, y) = \downarrow x \times \downarrow y$.*

Proof Is an immediate consequence of the following:

$$\begin{aligned} (a, b) \in \downarrow(x, y) &\Leftrightarrow (a, b) \leq (x, y) \Leftrightarrow a \leq_1 x \text{ and } b \leq_2 y \\ &\Leftrightarrow a \in \downarrow x \text{ and } b \in \downarrow y \Leftrightarrow (a, b) \in \downarrow x \times \downarrow y. \end{aligned}$$

□

Lemma 3 Let $(x, y) \in X \times Y$. Then, (x, y) is a dense element of $X \times Y$ if and only if x and y are dense elements of X and Y , respectively.

Proof First we observe that for every $(x_1, y_1), (x_2, y_2) \in X \times Y$ we have

$$(x_1, y_1) \wedge (x_2, y_2) = (x_1 \wedge x_2, y_1 \wedge y_2) \text{ and } (x_1, y_1) \vee (x_2, y_2) = (x_1 \vee x_2, y_1 \vee y_2).$$

Suppose that (x, y) is a dense element of $X \times Y$ and we prove that x is a dense element of X . Let $x' \in X \setminus \{0_X\}$. We consider the element $(x', 0_Y)$ of $X \times Y$. Then, $(x', 0_Y) \neq 0_{X \times Y}$ and since (x, y) is a dense element of $X \times Y$,

$$\begin{aligned} (x, y) \wedge (x', 0_Y) \neq 0_{X \times Y} &\Leftrightarrow (x \wedge x', y \wedge 0_Y) \neq (0_X, 0_Y) \\ &\Leftrightarrow (x \wedge x', 0_Y) \neq (0_X, 0_Y) \Leftrightarrow x \wedge x' \neq 0_X. \end{aligned}$$

Similarly we can prove that y is a dense element of Y .

Conversely, suppose that x and y are dense elements of X and Y , respectively. Let $(x', y') \in X \times Y \setminus \{0_{X \times Y}\}$. Then, $x' \neq 0_X$ or $y' \neq 0_Y$. Therefore, $x \wedge x' \neq 0_X$ or $y \wedge y' \neq 0_Y$. Thus, $(x, y) \wedge (x', y') = (x \wedge x', y \wedge y') \neq 0_{X \times Y}$ and the proof of the lemma is complete. $\square$

Proposition 6 Let X and Y be finite lattices. Then,

$$\dim_q(X \times Y) = \max\{\dim_q(X), \dim_q(Y)\}.$$

Proof By Proposition 1 we have

$$\dim_q(X \times Y) = \max\{\dim(\downarrow(x, y)) : (x, y) \text{ is a dense element in } X \times Y\}.$$

By Lemmas 2 and 3 we have

$$\dim_q(X \times Y) = \max\{\dim(\downarrow x \times \downarrow y) : x \text{ and } y \text{ are dense in } X \text{ and } Y\}$$

and by Zhang et al. (2017) we have

$$\dim_q(X \times Y) = \max\{\max\{\dim(\downarrow x), \dim(\downarrow y)\} : x \text{ and } y \text{ are dense in } X \text{ and } Y\}.$$

Thus, adapting Proposition 1 for the quasi covering dimension of X and Y we get

$$\dim_q(X \times Y) = \max\{\dim_q(X), \dim_q(Y)\}.$$

$\square$

Remark 5 (1) Let X and Y be finite lattices. Then, we obtain that $\dim_q(X \times Y) = 0$ if and only if $\dim_q(X) = \dim_q(Y) = 0$.

(2) If X and Y are finite lattices, then

$$\dim_q(X \times Y) < \dim_q(X) + \dim_q(Y) + 1.$$

(3) If $X_1, \dots, X_n$ are finite lattices, then

$$\dim_q(X_1 \times \dots \times X_n) = \max\{\dim_q(X_1), \dots, \dim_q(X_n)\}.$$

Definition 11 (Davey and Priestley 2002) The *lexicographic product* $X \diamond Y$ of two lattices $(X, \leq_1)$ and $(Y, \leq_2)$ is the lattice $(X \times Y, \leq)$, where the relation $\leq$ is defined as follows:

$$(x_1, y_1) \leq (x_2, y_2) \Leftrightarrow \begin{cases} x_1 <_1 x_2 \text{ or} \\ x_1 = x_2 \text{ and } y_1 \leq_2 y_2. \end{cases}$$

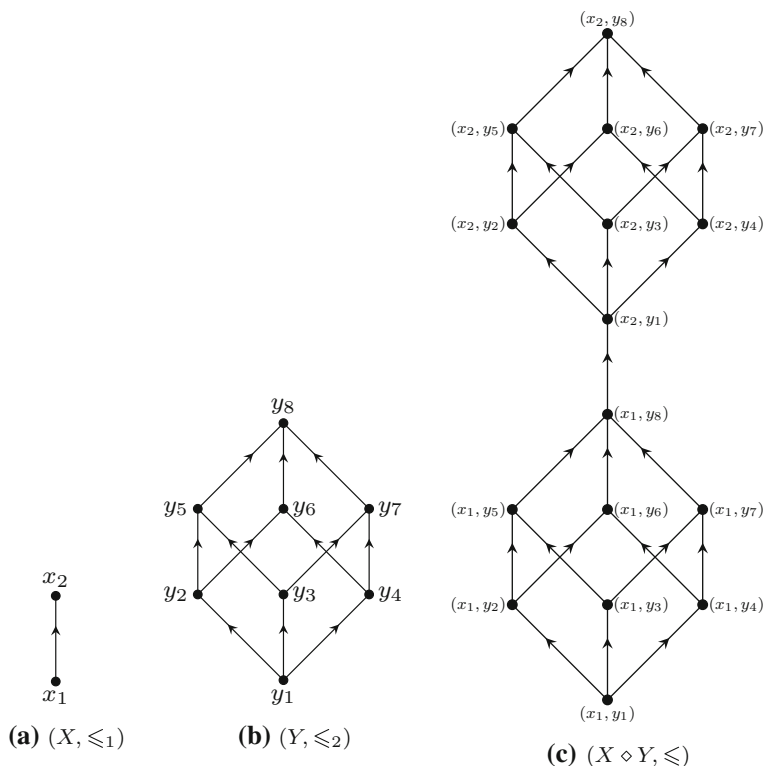


Fig. 8 The finite lattices $(X, \leq_1)$, $(Y, \leq_2)$ and $(X \diamond Y, \leq)$

Remark 6 The example in Fig. 8 verifies that the inequality

$$\dim_q(X \diamond Y) \leq \dim_q(X) + \dim_q(Y) + 1$$

does not hold for all the finite lattices X and Y .

Indeed, we have $\dim_q(X) = \dim_q(Y) = 0$ and $\dim_q(X \diamond Y) = 2$. Thus, the inequality $\dim_q(X \diamond Y) \leq \dim_q(X) + \dim_q(Y) + 1$ does not hold.

Definition 12 (Bae 1994, 1999; Bennett 1990) The *rectangular product* of two lattices $(X, \leq_1)$ and $(Y, \leq_2)$ is the lattice $(X \square Y, \leq)$, where

$$X \square Y = \{(x, y) \in X \times Y : x \neq x_1 \text{ and } y \neq y_1\} \cup \{(x_1, y_1)\}$$

and the relation $\leq$ is defined as follows:

$$(x_1, y_1) \leq (x_2, y_2) \Leftrightarrow x_1 \leq_1 x_2 \text{ and } y_1 \leq_2 y_2.$$

Remark 7 The example in Fig. 9 shows that the inequality

$$\dim_q(X \square Y) \leq \dim_q(X) + \dim_q(Y)$$

does not hold for all the finite lattices.

Indeed, we have $\dim_q(X) = \dim_q(Y) = 0$ and $\dim_q(X \square Y) = 1$. Thus, the inequality $\dim_q(X \square Y) \leq \dim_q(X) + \dim_q(Y)$ does not hold.

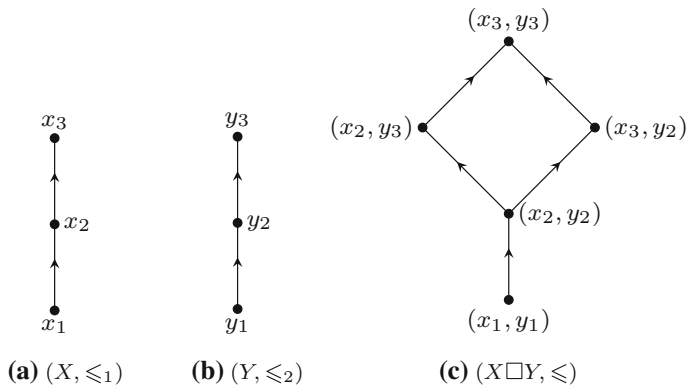


Fig. 9 The finite lattices $(X, \leq_1)$, $(Y, \leq_2)$ and $(X \square Y, \leq)$

5 A characterization of dense elements using matrices

In this section we characterize the dense elements of an arbitrary finite lattice $(X = \{x_1, \dots, x_n\}, \leq)$ using matrices.

Notation 1 (Boyadzhiev et al. 2018) We denote by $B_m = (b_m^i)$ the $n \times 1$ matrix, where

$$b_m^i = \begin{cases} m, & \text{if } i = 1 \\ 0, & \text{otherwise.} \end{cases}$$

Proposition 7 Let X be a finite lattice and $x_i \in X$, where $i \in \{2, \dots, n\}$ be an arbitrary element of X . Then, x_i is a dense element of X if and only if

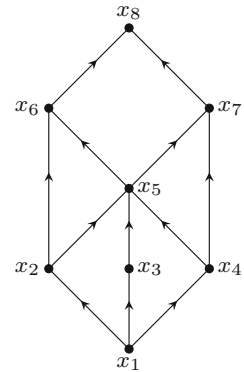
$$\max(c_i(T_X^{\leq}) + c_j(T_X^{\leq}) - B_1) = 2, \text{ for every } j \in \{2, \dots, n\} \setminus \{i\}.$$

Proof Let $x_i \in X$ be a dense element of X , where $i \in \{2, \dots, n\}$, $j \in \{2, \dots, n\} \setminus \{i\}$ and $k = \max(c_i(T_X^{\leq}) + c_j(T_X^{\leq}) - B_1)$. Clearly, $k > 0$ and by the corresponding definitions of the incidence matrix $T_X^{\leq}$ and the matrix B_1 we have either $k = 1$ or $k = 2$. Since x_i is a dense element of X , we have $x_i \wedge x_j \neq x_1$. We set $x_m = x_i \wedge x_j$, where $m \in \{2, \dots, n\}$. Then, $x_m \leq x_i$ and $x_m \leq x_j$, or equivalently, $t_{mi} = t_{mj} = 1$. Additionally, $x_m \neq x_1$. Thus, $b_1^m = 0$. Therefore, $t_{mi} + t_{mj} - b_1^m = 2$ and so $k = 2$.

Conversely, we suppose that $\max(c_i(T_X^{\leq}) + c_j(T_X^{\leq}) - B_1) = 2$, for every $j \in \{2, \dots, n\} \setminus \{i\}$ and we prove that the element x_i is dense in X . For that, we assume that the element x_i is not dense in X . Then, there exists $q \in \{2, \dots, n\} \setminus \{i\}$ such that $x_i \wedge x_q = x_1$. By assumption, $\max(c_i(T_X^{\leq}) + c_q(T_X^{\leq}) - B_1) = 2$. Thus, there exists $s \in \{2, \dots, n\}$ such that $t_{si} + t_{sq} - b_1^s = 2$. The last means that $t_{si} = t_{sq} = 1$ and $b_1^s = 0$, or equivalent $x_s \leq x_i$, $x_s \leq x_q$ and $x_s \neq x_1$. Therefore, $x_s < x_i \wedge x_q$ and so $x_s < x_1$, which is a contradiction. Thus, x_i is a dense element of X . $\square$

Algorithm 1 Let $(X = \{x_1, \dots, x_n\}, \leq)$ be a finite lattice. The algorithm which finds the set $D(X)$ of all dense elements of X is consisted by the following steps:

Step 1: Find the n columns $c_1(T_X^{\leq}), \dots, c_n(T_X^{\leq})$ of the incidence matrix $T_X^{\leq}$ of X and go to Step 2.

Fig. 10 The finite lattice $(X, \leq)$ 

Step 2: Set $k = 0$, $i = 2$ and go to Step 3.

Step 3: If $\max(c_i(T_X^{\leq}) + c_j(T_X^{\leq}) - B_1) = 2$, for every $j \in \{2, \dots, n\} \setminus \{i\}$, then put $x_i = x_{i_k}$ and go to step 4. Otherwise, go to Step 5.

Step 4: Put $k \rightarrow k + 1$ (that is, replace the old k with $k + 1$) and go to Step 5.

Step 5: If $i < n$, then put $i \rightarrow i + 1$ (that is, replace the old i with $i + 1$) and go to Step 3. Otherwise, go to Step 6.

Step 6: Create the set $D(X) = \{x_{i_0}, \dots, x_{i_k}\}$.

Example 3 Let $(X, \leq)$ be the finite lattice represented in Fig. 10.

The incidence matrix $T_X^{\leq}$ of X is the following matrix:

$$T_X^{\leq} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Following Algorithm 1 we have that the dense elements of X are the elements x_5 , x_6 , x_7 and x_8 . For example the element x_5 is dense in X since

$$c_5(T_X^{\leq}) + c_2(T_X^{\leq}) - B_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad c_5(T_X^{\leq}) + c_3(T_X^{\leq}) - B_1 = \begin{pmatrix} 1 \\ 1 \\ 2 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix},$$

$$\begin{aligned}
 c_5(T_X^{\leq}) + c_4(T_X^{\leq}) - B_1 &= \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad c_5(T_X^{\leq}) + c_6(T_X^{\leq}) - B_1 = \begin{pmatrix} 1 \\ 2 \\ 2 \\ 2 \\ 2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \\
 c_5(T_X^{\leq}) + c_7(T_X^{\leq}) - B_1 &= \begin{pmatrix} 1 \\ 2 \\ 2 \\ 2 \\ 2 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad c_5(T_X^{\leq}) + c_8(T_X^{\leq}) - B_1 = \begin{pmatrix} 1 \\ 2 \\ 2 \\ 2 \\ 2 \\ 1 \\ 1 \\ 1 \end{pmatrix}.
 \end{aligned}$$

Thus, $\max(c_5(T_X^{\leq}) + c_j(T_X^{\leq}) - B_1) = 2$, for every $j = 2, 3, 4, 6, 7, 8$. But the element x_3 is not dense in X since there exists $j = 2$ such that

$$c_3(T_X^{\leq}) + c_2(T_X^{\leq}) - B_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

Thus, $\max(c_3(T_X^{\leq}) + c_2(T_X^{\leq}) - B_1) = 1 \neq 2$.

6 An algorithm for computing the quasi covering dimension of finite lattices

In this section we present an algorithm for computing the quasi covering dimension of an arbitrary finite lattice $(X = \{x_1, \dots, x_n\}, \leq)$.

Definition 13 (Dube et al. 2015) A cover C of X is called *minimal* if $C \subseteq R$, for every refinement R of C . As we can see in Dube et al. (2015), every minimal cover is an antichain.

In the study Boyadzhiev et al. (2018), based on the following relation

$$\dim(X) = \max\{\text{ord}(C) : C \text{ is a minimal cover of } X\},$$

for every finite lattice X , which was proved in the study Dube et al. (2015), we gave two algorithms, each one of which computes the set $MCov(X)$ of all minimal covers of X and the covering dimension, $\dim(X)$, of X . For that, we inserted the following definition, proposition and notation, respectively.

Definition 14 (Boyadzhiev et al. 2018) Let X be a finite lattice.

- (1) An antichain subset $C = \{x_j\}$ of X is said to *satisfy the cover condition* if the following condition is satisfied:

$$3 \notin r_j(\mathbf{A}_X^{\leq}) + (-1) \cdot r_n(\mathbf{A}_X^{\leq}).$$

- (2) An antichain subset $C = \{x_{j_1}, \dots, x_{j_m}\}$, where $m \geq 2$, is said to *satisfy the cover condition* if the following condition is satisfied:

$$2m + 2 \notin r_{j_1}(\mathbf{A}_X^{\leq}) + \dots + r_{j_m}(\mathbf{A}_X^{\leq}) + (-1) \cdot r_n(\mathbf{A}_X^{\leq}).$$

Proposition 8 (Boyadzhiev et al. 2018) *Let X be a finite lattice.*

- (1) *An antichain subset $C = \{x_j\}$ of X is a cover of X if and only if it satisfies the cover condition.*
 (2) *An antichain subset $C = \{x_{j_1}, \dots, x_{j_m}\}$ of X , where $m \geq 2$, is a cover of X if and only if it satisfies the cover condition.*

Notation 2 (Boyadzhiev et al. 2018) If A is an antichain subset of X , then we write

$$Pl(A) = \{x \in X : x \parallel a \text{ for every } a \in A\}$$

and for every $x_i \in X$ we write

$$\downarrow^* x_i = \{x \in X : x \leq x_i\} \setminus \{0_X\}.$$

Algorithm 2 (Boyadzhiev et al. 2018) Let $(X = \{x_1, \dots, x_n\}, \leq)$ be a finite lattice, where $n \geq 2$. The algorithm which finds the set $MCov(X)$ of all minimal covers of X consists of the following steps.

Step 1: Find the n rows $r_1(\mathbf{A}_X^{\leq}), \dots, r_n(\mathbf{A}_X^{\leq})$ of the order matrix $\mathbf{A}_X^{\leq}$ of X and go to Step 2.

Step 2: If $0 \notin \mathcal{E}(\mathbf{A}_X^{\leq})$, then create $MCov(X) = \{\{x_n\}\}$. Otherwise, go to Step 3.

Step 3: Set $m = 1$ and go to Step 4.

Step 4: Find the set $\mathcal{A}_m(X)$ of all antichains $\{x_{j_1}, \dots, x_{j_m}\}$ of X which satisfy the cover condition and go to Step 5.

Step 5: If $m < n - 2$, then put $m \rightarrow m + 1$ (that is, replace the old m with $m + 1$) and go to Step 4. Otherwise, go to Step 6.

Step 6: If $\mathcal{A}_m(X) = \emptyset$, for all $m = 2, \dots, n - 2$, then create $MCov(X) = \{\{x_n\}\}$. Otherwise, go to Step 7.

Step 7: Create the set $\mathcal{A}(X) = \mathcal{A}_1(X) \cup \dots \cup \mathcal{A}_{n-2}(X)$ and go to Step 8.

Step 8: Find the set $\mathcal{M}(X)$ of all sets $C = \{x_{j_1}, \dots, x_{j_m}\} \in \mathcal{A}(X)$ such that for every $i \in \{1, \dots, m\}$ the following conditions hold:

- (1) $C \setminus \{x_{j_i}\}$ does not satisfy the cover condition.
- (2) If for every $k_1, k_2 \in \{1, \dots, n\} \setminus \{j_1, \dots, j_m\}$, $1 \in r_{k_1}(\mathbf{A}_X^{\leq}) + r_{k_2}(\mathbf{A}_X^{\leq})$, $3 \in r_{j_i}(\mathbf{A}_X^{\leq}) + r_{k_1}(\mathbf{A}_X^{\leq})$, $3 \in r_{j_i}(\mathbf{A}_X^{\leq}) + r_{k_2}(\mathbf{A}_X^{\leq})$, $1 \in r_{k_1}(\mathbf{A}_X^{\leq}) + r_{j_i}(\mathbf{A}_X^{\leq})$, $1 \in r_{k_2}(\mathbf{A}_X^{\leq}) + r_{j_i}(\mathbf{A}_X^{\leq})$, for every $l \in \{1, \dots, m\} \setminus \{i\}$, then $(C \setminus \{x_{j_i}\}) \cup \{x_{k_1}, x_{k_2}\}$ does not satisfy the cover condition.
- (3) If for every $k \in \{1, \dots, n\} \setminus \{j_1, \dots, j_m\}$, $3 \in r_{j_i}(\mathbf{A}_X^{\leq}) + r_k(\mathbf{A}_X^{\leq})$ and $1 \in r_k(\mathbf{A}_X^{\leq}) + r_{j_i}(\mathbf{A}_X^{\leq})$, for every $l \in \{1, \dots, m\} \setminus \{i\}$, then $(C \setminus \{x_{j_i}\}) \cup \{x_k\}$ does not satisfy the cover condition.

Subsequently, go to Step 9.

Step 9: Create the set $MCov(X) = \mathcal{M}(X)$.

Algorithm 3 (Boyadzhiev et al. 2018) Let $(X = \{x_1, \dots, x_n\}, \leq)$ be a finite lattice.

Step 1: Find the n rows $r_1(\mathbf{A}_X^{\leq}), \dots, r_n(\mathbf{A}_X^{\leq})$ of the order matrix $\mathbf{A}_X^{\leq}$ of X . If $0 \notin \mathcal{E}(\mathbf{A}_X^{\leq})$, then print $\dim(X) = 0$. Otherwise, go to Step 2.

Step 2: Apply Algorithm 2 to find the set $\mathcal{A}(X) = \bigcup_{m=1}^{n-2} \mathcal{A}_m(X)$ and go to Step 3.

Step 3: If $\mathcal{A}_m(X) = \emptyset$, for all $m = 2, \dots, n-2$, then print $\dim(X) = 0$. Otherwise, go to Step 4.

Step 4: Apply Algorithm 2 to find the set $\mathcal{M}(X)$ and go to Step 5.

Step 5: Find the n columns $c_1(T_X^{\leq}), \dots, c_n(T_X^{\leq})$ of the incidence matrix $T_X^{\leq}$ of X and go to Step 6.

Step 6: For each $m \in \{1, \dots, n-2\}$, find

$$d_m = \max\{\max(c_{j_1}(T_X^{\leq}) + \dots + c_{j_m}(T_X^{\leq}) - B_m) - 1 : \{x_{j_1}, \dots, x_{j_m}\} \in \mathcal{M}(X)\}.$$

If $\mathcal{A}_m(X) \cap \mathcal{M}(X) = \emptyset$, for some m , then put $d_m = 0$. Go to Step 7.

Step 7: Print $\dim(X) = \max\{d_1, d_2, \dots, d_{n-2}\}$.

Proposition 9 Let X be a finite lattice and x be a dense element of X .

(1) If

$$L_x = \max\{\max(c_{j_1}(T_X^{\leq}) + \dots + c_{j_m}(T_X^{\leq}) - B_m) - 1 : \{x_{j_1}, \dots, x_{j_m}\} \in \mathcal{M}(\downarrow x)\},$$

then $\dim(\downarrow x) = L_x$.

(2) If $Q = \max\{L_x : x \text{ is a dense element of } X\}$, then $\dim_q(X) = Q$.

Proof (1) It similar to the proof of Boyadzhiev et al. (2018, Theorem 5.2).

(2) It follows by Proposition 1 and the statement (1) of this proposition. $\square$

Adapting Algorithms 1, 2, and 3 we can compute the covering dimension $\dim(\downarrow x)$ for every dense element x of an arbitrary finite lattice X . According to Proposition 1, this leads to a computation of the quasi covering dimension, $\dim_q(X)$, of X .

Algorithm 4 Let $(X = \{x_1, \dots, x_n\}, \leq)$ be a finite lattice.

Step 1: Apply Algorithm 1 to find the set $D(X) = \{x_{i_0}, \dots, x_{i_k}\}$ of all dense elements of X .

Step 2: Set $s = 0$ and go to Step 3.

Step 3: Find the set $\downarrow x_{i_s} = \{x_j : a_{ji_s} \in \{1, 2\}\}$ and go to Step 4.

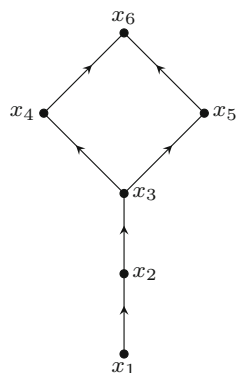
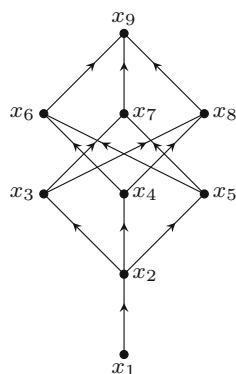
Step 4: Apply Algorithm 3 to find the $q_s = \dim(\downarrow x_{i_s})$ and go to Step 5.

Step 5: If $s < k$, then put $s \rightarrow s + 1$ (that is, replace the old s with $s + 1$) and go to Step 3. Otherwise, go to Step 6.

Step 6: Print $\dim_q(X) = \max\{q_0, q_1, \dots, q_k\}$.

Example 4 (1) We consider the finite lattice $(X, \leq)$ which is represented in Fig. 11.

Following Algorithm 1 we have $D(X) = \{x_2, x_3, x_4, x_5, x_6\}$. Additionally, following Algorithms 2 and 3 we have $\mathcal{M}(\downarrow x_2) = \{\{x_2\}\}$ with $q_0 = \dim(\downarrow x_2) = 0$, $\mathcal{M}(\downarrow x_3) = \{\{x_3\}\}$ with $q_1 = \dim(\downarrow x_3) = 0$, $\mathcal{M}(\downarrow x_4) = \{\{x_4\}\}$ with $q_2 = \dim(\downarrow x_4) = 0$, $\mathcal{M}(\downarrow x_5) = \{\{x_5\}\}$ with $q_3 = \dim(\downarrow x_5) = 0$ and $\mathcal{M}(\downarrow x_6) = \{\{x_4, x_5\}\}$ with $q_4 = \dim(\downarrow x_6) = 1$. Thus, $\dim_q(X) = \max\{q_0, q_1, q_2, q_3, q_4\} = 1$.

Fig. 11 The finite lattice $(X, \leq)$ **Fig. 12** The finite lattice $(X, \leq)$ 

(2) We consider the finite lattice $(X, \leq)$ which is represented in Fig. 12.

Following Algorithm 1 we have $D(X) = \{x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9\}$ and following Algorithms 2 and 3 we have $\mathcal{M}(\downarrow x_i) = \{\{x_i\}\}$ and $q_{i-2} = \dim(\downarrow x_i) = 0$, for every $i = 2, 3, 4, 5$, $\mathcal{M}(\downarrow x_6) = \{\{x_4, x_5\}\}$ with $q_4 = \dim(\downarrow x_6) = 1$, $\mathcal{M}(\downarrow x_7) = \{\{x_3, x_5\}\}$ with $q_5 = \dim(\downarrow x_7) = 1$, $\mathcal{M}(\downarrow x_8) = \{\{x_3, x_4\}\}$ with $q_6 = \dim(\downarrow x_8) = 1$ and $\mathcal{M}(\downarrow x_9) = \{\{x_3, x_4, x_5\}\}$ with $q_7 = \dim(\downarrow x_9) = 2$. Therefore, $\dim_q(X) = \max\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7\} = 2$.

Conclusion

In this paper, we insert a new dimension for finite lattices, called the quasi covering dimension. We study many properties of this dimension and we characterize it using matrices. An algorithm for computing the quasi covering dimension of an arbitrary finite lattice completes our study.

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